Lemma 9. *If we use stepsize $N\tilde{\eta_V}$ for $\boldsymbol{W}_V$, then it holds that*

$$\langle \boldsymbol{e}_k, \boldsymbol{W}_V^{*,(1)} \boldsymbol{p}_t \rangle = \begin{cases} -\alpha_t \tilde{\eta_V} & (k \in [N_{\mathrm{trg}}]), \\ \frac{\alpha_t \tilde{\eta_V} N_{\mathrm{trg}}}{N - N_{\mathrm{trg}}} & (k \notin [N_{\mathrm{trg}}]), \end{cases} \tag{B.1}$$

$$\langle \boldsymbol{e}_j, \boldsymbol{W}_V^{*,(2)} \boldsymbol{e}_k \rangle = \begin{cases} -2\tilde{\eta_V} \mathbb{E}[T^{-1}] N_{\mathrm{trg}}^{-1} & (j, k \in [N_{\mathrm{trg}}]) \\ -\tilde{\eta_V} \frac{1 - 2\mathbb{E}[T^{-1}]}{N - N_{\mathrm{trg}}} & (j \in [N_{\mathrm{trg}}], k \notin [N_{\mathrm{trg}}]) \\ 2\tilde{\eta_V} \frac{\mathbb{E}[T^{-1}]}{N - N_{\mathrm{trg}}} & (j \notin [N_{\mathrm{trg}}], k \in [N_{\mathrm{trg}}]) \\ \tilde{\eta_V} \frac{N_{\mathrm{trg}} + \mathbb{E}[T^{-1}](N(N-1) - N_{\mathrm{trg}}(N+2))}{(N - N_{\mathrm{trg}})^2} & (j = k \notin [N_{\mathrm{trg}}]) \\ \tilde{\eta_V} \frac{N_{\mathrm{trg}} - \mathbb{E}[T^{-1}](N + 2N_{\mathrm{trg}})}{(N - N_{\mathrm{trg}})^2} & (j \neq k,\ j, k \notin [N_{\mathrm{trg}}]) \end{cases} \tag{B.2}$$

and

$$\langle \boldsymbol{e}_j, \boldsymbol{W}_V^{*,(3)} \boldsymbol{e}_k \rangle = \begin{cases} -\tilde{\eta_V} \mathbb{E}[T^{-1}] N_{\mathrm{trg}}^{-1} & (j, k \in [N_{\mathrm{trg}}]) \\ -\tilde{\eta_V} \frac{1 - 2\mathbb{E}[T^{-1}]}{N - N_{\mathrm{trg}}} & (j \in [N_{\mathrm{trg}}], k \notin [N_{\mathrm{trg}}]) \\ \tilde{\eta_V} \frac{\mathbb{E}[T^{-1}]}{N - N_{\mathrm{trg}}} & (j \notin [N_{\mathrm{trg}}], k \in [N_{\mathrm{trg}}]) \\ \tilde{\eta_V} \frac{N_{\mathrm{trg}} + \mathbb{E}[T^{-1}](N(N-1) - N_{\mathrm{trg}}(N+2))}{(N - N_{\mathrm{trg}})^2} & (j = k \notin [N_{\mathrm{trg}}]) \\ \tilde{\eta_V} \frac{N_{\mathrm{trg}} - \mathbb{E}[T^{-1}](N + 2N_{\mathrm{trg}})}{(N - N_{\mathrm{trg}})^2} & (j \neq k,\ j, k \notin [N_{\mathrm{trg}}]), \end{cases}$$

where we defined $\alpha_t \coloneqq \left\langle \mathbb{E}_T\left[T^{-1} \mathbf{1}_{1:T}\right], \boldsymbol{p}_t \right\rangle$: specifically, $\alpha_t = \mathbb{E}_T[T^{-1} \mathbb{I}\{t \leq T\}] \leq \mathbb{E}_T[T^{-1}]$ is satisfied.

Proof.

For the first block, we have the following evaluation of the gradient of population loss $\bar{\mathcal{L}}$:

$$\begin{aligned} &\nabla_{\boldsymbol{W}_V^{(1)}} \bar{\mathcal{L}}(\boldsymbol{W}_{KQ}, \boldsymbol{W}_V) \\ =&\frac{1}{N} \sum_{k=1}^{N} \boldsymbol{e}_k \mathbb{E}_T \left[\frac{1}{T} \sum_{t=1}^{T} \mathbb{E}[\boldsymbol{p}_t]^\top \right] - \frac{1}{N - N_{\mathrm{trg}}} \sum_{k=N_{\mathrm{trg}}+1}^{N} \boldsymbol{e}_k \mathbb{E}_T \left[\frac{1}{T} \sum_{t=1}^{T} \mathbb{E}[\boldsymbol{p}_t | z_{T+1} = k]^\top \right] \\ =&\frac{1}{N} \sum_{k=1}^{N} \boldsymbol{e}_k \mathbb{E}_T \left[T^{-1} \sum_{t=1}^{T} \boldsymbol{p}_t^\top \right] - \frac{1}{N - N_{\mathrm{trg}}} \sum_{k=N_{\mathrm{trg}}+1}^{N} \boldsymbol{e}_k \mathbb{E}_T \left[T^{-1} \sum_{t=1}^{T} \boldsymbol{p}_t^\top \right]. \end{aligned}$$

This immediately yields (B.1).

Now for the second block, we have

$$\begin{aligned} &\nabla_{\boldsymbol{W}_V^{(2)}} \bar{\mathcal{L}}(\boldsymbol{W}_{KQ}, \boldsymbol{W}_V) \\ =&\frac{1}{N} \sum_{k=1}^{N} \boldsymbol{e}_k \mathbb{E}_T \left[\frac{1}{T} \sum_{t=1}^{T} \mathbb{E}[\boldsymbol{e}_{z_t}]^\top \right] - \frac{1}{N - N_{\mathrm{trg}}} \sum_{k=N_{\mathrm{trg}}+1}^{N} \boldsymbol{e}_k \mathbb{E}_T \left[\frac{1}{T} \sum_{t=1}^{T} \mathbb{E}[\boldsymbol{e}_{z_t} | z_{T+1} = k]^\top \right], \end{aligned}$$

where the first term is evaluated as

$$\frac{1}{N} \sum_{k=1}^{N} \boldsymbol{e}_k \mathbb{E}_T \left[\frac{1}{T} \sum_{t=1}^{T} \mathbb{E}[\boldsymbol{e}_{z_t}]^\top \right] = \frac{1}{N} \sum_{k=1}^{N} \boldsymbol{e}_k \sum_{w=1}^{N_{\mathrm{trg}}} \frac{1}{N_{\mathrm{trg}}} \mathbb{E}_T \left[\frac{1}{T} \left(2\boldsymbol{e}_w + \frac{T-2}{N - N_{\mathrm{trg}}} \mathbf{1}_{N_{\mathrm{trg}}+1:N} \right)^\top \right],$$

and similarly for the second term

$$\begin{aligned} &\frac{1}{N - N_{\mathrm{trg}}} \sum_{k=N_{\mathrm{trg}}+1}^{N} \boldsymbol{e}_k \mathbb{E}_T \left[\frac{1}{T} \sum_{t=1}^{T} \mathbb{E}[\boldsymbol{e}_{z_t} | z_{T+1} = k]^\top \right] \\ =&\frac{1}{N - N_{\mathrm{trg}}} \sum_{k=N_{\mathrm{trg}}+1}^{N} \boldsymbol{e}_k \frac{1}{N_{\mathrm{trg}}} \sum_{w=1}^{N_{\mathrm{trg}}} \mathbb{E}_T \left[\frac{1}{T} \left(2\boldsymbol{e}_w + \frac{T-3}{N - N_{\mathrm{trg}}} \mathbf{1}_{N_{\mathrm{trg}}+1:N} + \boldsymbol{e}_k \right)^\top \right]. \end{aligned}$$

Putting everything together yields (B.2).

The third block can be computed in the same fashion as

$$\begin{aligned}&\nabla_{\boldsymbol{W}_V^{(3)}}\bar{\mathcal{L}}(\boldsymbol{W}_{KQ},\boldsymbol{W}_V)\\=&\frac{1}{N}\sum_{k=1}^{N}\boldsymbol{e}_k\mathbb{E}_T\left[\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}[\boldsymbol{e}_{z_{t-1}}]^\top\right]-\frac{1}{N-N_{\mathrm{trg}}}\sum_{k=N_{\mathrm{trg}}+1}^{N}\boldsymbol{e}_k\mathbb{E}_T\left[\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}[\boldsymbol{e}_{z_{t-1}}|z_{T+1}=k]^\top\right],\end{aligned}$$

where we have the evaluations

$$\begin{aligned}&\frac{1}{N}\sum_{k=1}^{N}\boldsymbol{e}_k\mathbb{E}_T\left[\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}[\boldsymbol{e}_{z_{t-1}}]^\top\right]\\=&\frac{1}{N}\sum_{k=1}^{N}\boldsymbol{e}_k\frac{1}{N_{\mathrm{trg}}}\sum_{w=1}^{N_{\mathrm{trg}}}\mathbb{E}_T\left[\frac{1}{T}\left(\boldsymbol{e}_w+\frac{T-2}{N-N_{\mathrm{trg}}}\mathbf{1}_{N_{\mathrm{trg}}+1:N}\right)^\top\right]\ (\because \boldsymbol{e}_0=\mathbf{0})\end{aligned}$$

and

$$\begin{aligned}&\frac{1}{N-N_{\mathrm{trg}}}\sum_{k=N_{\mathrm{trg}}+1}^{N}\boldsymbol{e}_k\mathbb{E}_T\left[\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}[\boldsymbol{e}_{z_{t-1}}|z_{T+1}=k]^\top\right]\\=&\frac{1}{N-N_{\mathrm{trg}}}\sum_{k=N_{\mathrm{trg}}+1}^{N}\boldsymbol{e}_k\frac{1}{N_{\mathrm{trg}}}\sum_{w=1}^{N_{\mathrm{trg}}}\mathbb{E}_T\left[\frac{1}{T}\left(\boldsymbol{e}_w+\frac{T-3}{N-N_{\mathrm{trg}}}\mathbf{1}_{N_{\mathrm{trg}}+1:N}+\boldsymbol{e}_k\right)^\top\right].\end{aligned}$$

□

B.2 Population Gradient of $\mathbf{W}_{KQ}$

B.2.1 Preparations

We denote the transformer's predicted probability of token k given an input sequence z after one-step GD on $\boldsymbol{W}_V$ as

$$\hat{p}(k|z)=\mathrm{Softmax}(f_{\mathrm{TF}}(\boldsymbol{X}_{1:T};\boldsymbol{W}_{KQ},\boldsymbol{W}_V^*))_k.$$

We can approximate $\hat{p}(k|z)$ by considering sufficiently small $\tilde{\eta_V}$: The following corollary is obtained by Lemma 9.

Corollary 10. *If we set $\tilde{\eta_V}\lesssim 1/N$, then $|1/N-\hat{p}(k|z)|=O(\mathbb{E}[T^{-1}]/N^2)$ uniformly holds for any k and z.*

Proof. From Lemma 9, it holds that $|(f_{\mathrm{TF}}(\boldsymbol{X}_{1:T};\boldsymbol{W}_{KQ},\boldsymbol{W}_V^*))_k|\in O(\mathbb{E}[T^{-1}]/N+1/N^2)\in O(\mathbb{E}[T^{-1}]/N)$ ($\because$ Assumption 1) for all k. If $v_i=O(\mathbb{E}[T^{-1}]/N)$ for all $i\in[N]$ where $\boldsymbol{v}\in\mathbb{R}^N$, then it holds that

$$\frac{1}{N}\exp\bigl(-O(\mathbb{E}[T^{-1}]/N)\bigr)\le\mathrm{Softmax}(\boldsymbol{v})_i\le\frac{1}{N}\exp\bigl(O(\mathbb{E}[T^{-1}]/N)\bigr).$$

From Taylor's theorem we obtain the assertion. □

Following [BCB+23, Lemma 4], starting from $\boldsymbol{W}_{KQ}=\mathbf{0}$,

$$\begin{aligned}&\nabla_{\boldsymbol{W}_{KQ}}\bar{\mathcal{L}}(\boldsymbol{W}_{KQ},\boldsymbol{W}_V^*)\\=&\mathbb{E}\left[\sum_{k=1}^{N}(\hat{p}(k|z)-\mathbb{I}\{z_{T+1}=k\})\nabla_{\boldsymbol{W}_{KQ}=\mathbf{0}}\bigl\langle\boldsymbol{e}_k,\boldsymbol{W}_V^*\boldsymbol{X}_{1:T}\mathrm{Softmax}(\boldsymbol{X}_{1:T}^\top\boldsymbol{W}_{KQ}\boldsymbol{x}_T)\bigr\rangle\right],\end{aligned}$$

where

$$\nabla_{\boldsymbol{W}_{KQ}=\mathbf{0}}\left(\boldsymbol{e}_k^\top\boldsymbol{W}_V^*\sum_{t=1}^{T}\boldsymbol{x}_t\mathrm{Softmax}(\boldsymbol{X}_{1:T}^\top\boldsymbol{W}_{KQ}\boldsymbol{x}_T)_t\right)$$

$$
\begin{aligned}
&= \sum_{t=1}^{T} \boldsymbol{e}_k^\top \boldsymbol{W}_V^* \boldsymbol{x}_t \cdot \nabla_{\boldsymbol{W}_{KQ}=\boldsymbol{0}} \mathrm{Softmax}(\boldsymbol{X}_{1:T}^\top \boldsymbol{W}_{KQ} \boldsymbol{x}_T)_t \\
&= \frac{1}{T} \sum_{t=1}^{T} \boldsymbol{e}_k^\top \boldsymbol{W}_V^* \boldsymbol{x}_t \cdot (\boldsymbol{x}_t - \bar{\boldsymbol{x}}_{1:T}) \boldsymbol{x}_T^\top,
\end{aligned}
$$

for $\bar{\boldsymbol{x}}_{1:T} = \frac{1}{T}\sum_{t=1}^{T} \boldsymbol{x}_t$. Hence the population gradient simplifies to, assuming $\tilde{\eta_V} \lesssim 1/N$,

$$
\begin{aligned}
&\nabla_{\boldsymbol{W}_{KQ}} \bar{\mathcal{L}}(\boldsymbol{W}_{KQ}, \boldsymbol{W}_V^*) \\
&= \sum_{k=1}^{N} \mathbb{E}_T \left[\frac{1}{T} \sum_{t=1}^{T} \mathbb{E}_{\boldsymbol{X}} \left[\hat{p}(k|z) \boldsymbol{e}_k^\top \boldsymbol{W}_V^* \boldsymbol{x}_t \cdot (\boldsymbol{x}_t - \bar{\boldsymbol{x}}_{1:T}) \boldsymbol{x}_T^\top \right] \right] \\
&\quad - \sum_{k=N_{\mathrm{trg}}+1}^{N} \mathbb{E}_T \left[\frac{1}{T} \sum_{t=1}^{T} p(z_{T+1} = k) \mathbb{E}_{\boldsymbol{X}} \left[\boldsymbol{e}_k^\top \boldsymbol{W}_V^* \boldsymbol{x}_t \cdot (\boldsymbol{x}_t - \bar{\boldsymbol{x}}_{1:T}) \boldsymbol{x}_T^\top | z_{T+1} = k \right] \right] \\
&= \frac{1}{N} \sum_{k=1}^{N} \mathbb{E}_T \left[\frac{1}{T} \sum_{t=1}^{T} \mathbb{E}_{\boldsymbol{X}} \left[\boldsymbol{e}_k^\top \boldsymbol{W}_V^* \boldsymbol{x}_t \cdot (\boldsymbol{x}_t - \bar{\boldsymbol{x}}_{1:T}) \boldsymbol{x}_T^\top \right] \right] \\
&\quad - \frac{1}{N - N_{\mathrm{trg}}} \sum_{k=N_{\mathrm{trg}}+1}^{N} \mathbb{E}_T \left[\frac{1}{T} \sum_{t=1}^{T} \mathbb{E}_{\boldsymbol{X}} \left[\boldsymbol{e}_k^\top \boldsymbol{W}_V^* \boldsymbol{x}_t \cdot (\boldsymbol{x}_t - \bar{\boldsymbol{x}}_{1:T}) \boldsymbol{x}_T^\top | z_{T+1} = k \right] \right] \\
&\quad \underbrace{+ \sum_{k=1}^{N} \mathbb{E}_T \left[\frac{1}{T} \sum_{t=1}^{T} \mathbb{E}_{\boldsymbol{X}} \left[\left(\hat{p}(k|z) - \frac{1}{N} \right) \boldsymbol{e}_k^\top \boldsymbol{W}_V^* \boldsymbol{x}_t \cdot (\boldsymbol{x}_t - \bar{\boldsymbol{x}}_{1:T}) \boldsymbol{x}_T^\top \right] \right]}_{(*1)}.
\end{aligned}
$$

Note that each entry of $\boldsymbol{e}_k^\top \boldsymbol{W}_V^* \boldsymbol{x}_t \cdot (\boldsymbol{x}_t - \bar{\boldsymbol{x}}_{1:T}) \boldsymbol{x}_T^\top$ is of $O(\tilde{\eta_V} \mathbb{E}[T^{-1}])$ from Lemma 9, and $|\hat{p}(k|z) - \frac{1}{N}| \lesssim 1/N^2$ from Corollary 10. Therefore, using $|\mathbb{E}[ab]| \le \sqrt{\mathbb{E}[a^2 b^2]}$ we can conclude that (*1) is of $O_\infty(\tilde{\eta_V} \mathbb{E}[T^{-1}]/N)$.

Moreover, we have

$$
\begin{aligned}
&\frac{1}{N} \sum_{k=1}^{N} \mathbb{E}_T \left[\frac{1}{T} \sum_{t=1}^{T} \mathbb{E}_{\boldsymbol{X}} \left[\boldsymbol{e}_k^\top \boldsymbol{W}_V^* \boldsymbol{x}_t \cdot (\boldsymbol{x}_t - \bar{\boldsymbol{x}}_{1:T}) \boldsymbol{x}_T^\top \right] \right] \\
&= \frac{1}{N - N_{\mathrm{trg}}} \sum_{k=N_{\mathrm{trg}}+1}^{N} \mathbb{E}_T \left[\frac{1}{T} \sum_{t=1}^{T} \mathbb{E}_{\boldsymbol{X}} \left[\boldsymbol{e}_k^\top \boldsymbol{W}_V^* \boldsymbol{x}_t \cdot (\boldsymbol{x}_t - \bar{\boldsymbol{x}}_{1:T}) \boldsymbol{x}_T^\top \right] \right] \\
&\quad + \frac{1}{N - N_{\mathrm{trg}}} \sum_{k=1}^{N_{\mathrm{trg}}} \mathbb{E}_T \left[\frac{1}{T} \sum_{t=1}^{T} \mathbb{E}_{\boldsymbol{X}} \left[\boldsymbol{e}_k^\top \boldsymbol{W}_V^* \boldsymbol{x}_t \cdot (\boldsymbol{x}_t - \bar{\boldsymbol{x}}_{1:T}) \boldsymbol{x}_T^\top \right] \right] \\
&\quad - \frac{N_{\mathrm{trg}}}{N(N - N_{\mathrm{trg}})} \sum_{k=1}^{N} \mathbb{E}_T \left[\frac{1}{T} \sum_{t=1}^{T} \mathbb{E}_{\boldsymbol{X}} \left[\boldsymbol{e}_k^\top \boldsymbol{W}_V^* \boldsymbol{x}_t \cdot (\boldsymbol{x}_t - \bar{\boldsymbol{x}}_{1:T}) \boldsymbol{x}_T^\top \right] \right].
\end{aligned}
$$

Here, again using the fact that each entry of $\boldsymbol{e}_k^\top \boldsymbol{W}_V^* \boldsymbol{x}_t \cdot (\boldsymbol{x}_t - \bar{\boldsymbol{x}}_{1:T}) \boldsymbol{x}_T^\top$ is of $O(\tilde{\eta_V} \mathbb{E}[T^{-1}])$, the second and third terms can be bounded by $O_\infty(\tilde{\eta_V} \mathbb{E}[T^{-1}] \cdot N_{\mathrm{trg}}/N)$.

In conclusion, we obtain

$$
\begin{aligned}
&\nabla_{\boldsymbol{W}_{KQ}} \bar{\mathcal{L}}(\boldsymbol{W}_{KQ}, \boldsymbol{W}_V^*) \\
&= \frac{1}{N - N_{\mathrm{trg}}} \sum_{k=N_{\mathrm{trg}}+1}^{N} \mathbb{E}_T \left[\frac{1}{T} \sum_{t=1}^{T} \mathbb{E}_{\boldsymbol{X}} \left[\boldsymbol{e}_k^\top \boldsymbol{W}_V^* \boldsymbol{x}_t \cdot (\boldsymbol{x}_t - \bar{\boldsymbol{x}}_{1:T}) \boldsymbol{x}_T^\top \right] \right] \\
&\quad - \frac{1}{N - N_{\mathrm{trg}}} \sum_{k=N_{\mathrm{trg}}+1}^{N} \mathbb{E}_T \left[\frac{1}{T} \sum_{t=1}^{T} \mathbb{E}_{\boldsymbol{X}} \left[\boldsymbol{e}_k^\top \boldsymbol{W}_V^* \boldsymbol{x}_t \cdot (\boldsymbol{x}_t - \bar{\boldsymbol{x}}_{1:T}) \boldsymbol{x}_T^\top | z_{T+1} = k \right] \right]
\end{aligned}
$$